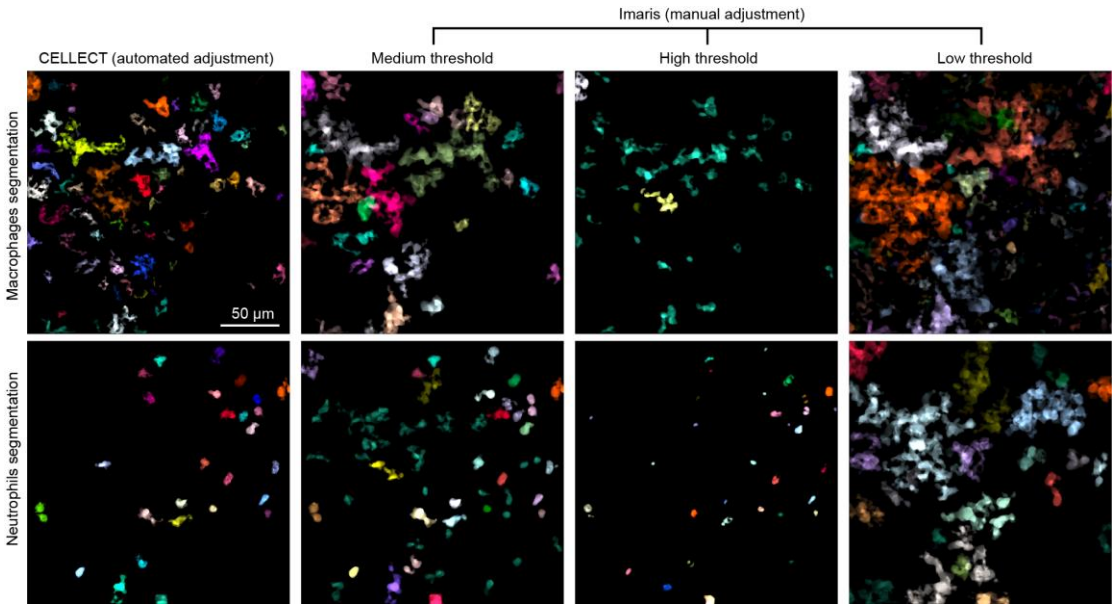


Supplementary Figure 19 | Influence of the intensity threshold on the segmentation results of Imaris with comparisons to CELLECT. The segmentation via CELLECT does not require threshold adjustment (left), ensuring more consistent and unbiased results interpretation. In contrast, the segmentation performance of the signal intensity-based algorithm in Imaris is highly sensitive to thresholding parameters, which can lead to variable outcomes from the same dataset. The data used here were acquired via 2pSAM imaging of neutrophils, macrophages, and bacteria in mouse spleen, consistent with the dataset shown in **Fig. 4**. CELLECT results were generated using the model trained on sequence #3 of the mskcc-confocal dataset. Scale bars: 50 µm.